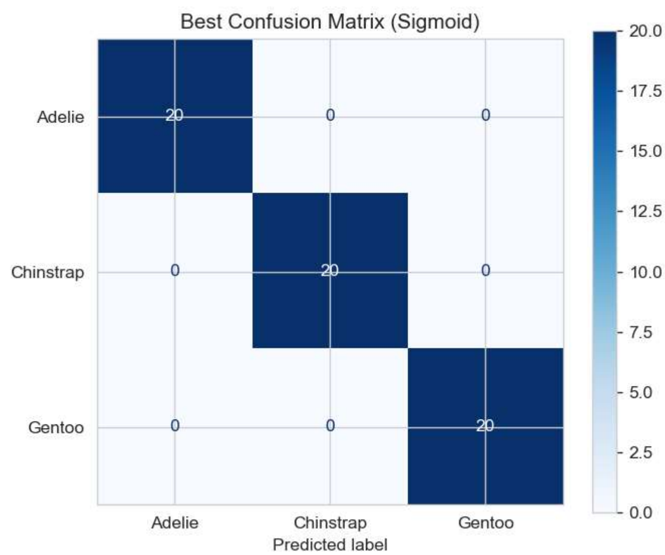
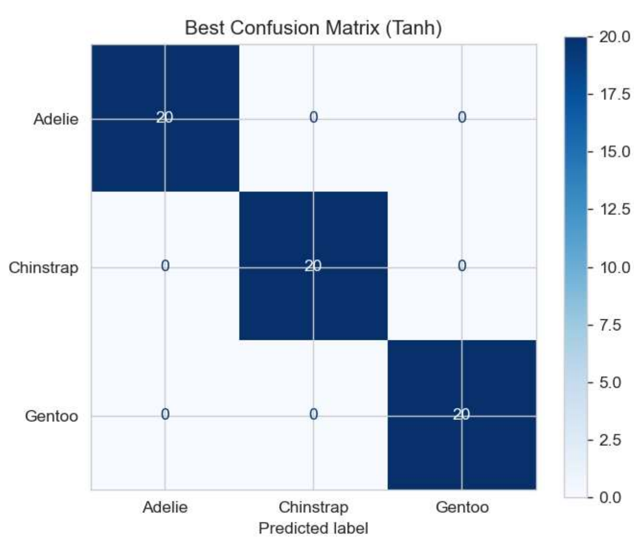
Best Model Parameters per Activation Function

Activation Function	Train Accuracy	Test Accuracy	LR	Epochs	#Layers	#HiddenNodes
sigmoid	100.00%	100.00%	0.01	500	1	[5]
tanh	100.00%	100.00%	0.01	500	1	[8]

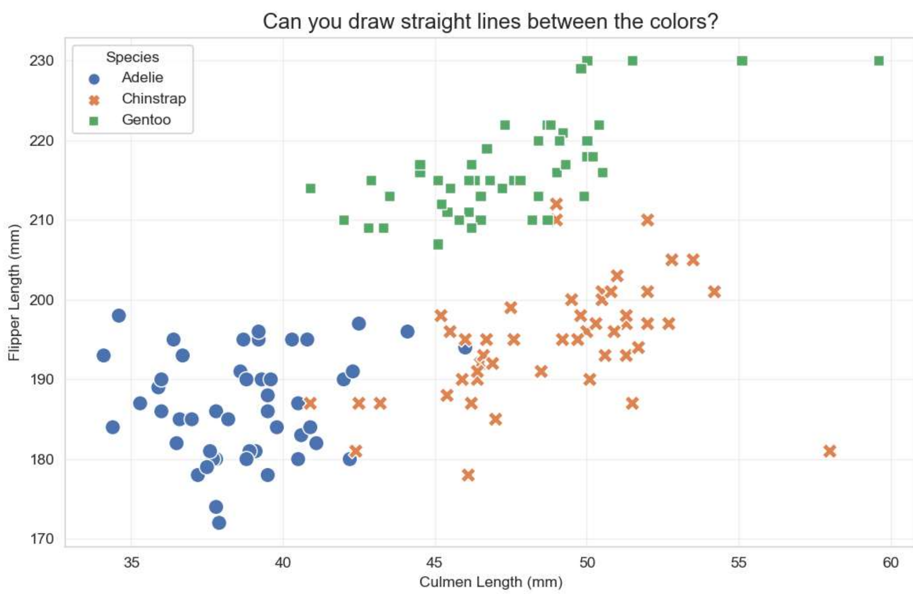




Observation:

Getting **100% accuracy** on both training and testing with a **1-layer** architecture tells us that no better accuracy with more complex architecture. Regarding models training, if simple model (1 layer) and complex model (2 layers) both give the same result, **always choose the simple one**, so we followed that rule and kept the 1-layer architecture as the more complex the model is, the slower, prone to overfitting it is and uses more memory.

The Palmer Penguins dataset is known to be a relatively "clean" dataset, which is linearly separable:



A single layer is enough to draw straight lines (decision boundaries) between these clusters. It doesn't need deep layers to learn complex curves because the data isn't complex.

We do not need complex hidden layers ("Deep Learning") for this specific dataset.

Easy puzzle!